Superposition and interference

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Add signed displacements and distinguish interference from lost energy.

SUCCESS CRITERIA - I can...

1. Add wave displacements at the same place and time.
2. Recognize constructive and destructive interference.
3. Explain that pulses continue after crossing in a linear medium.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “superposition” and “interference” to compare your model or result. Sentence starters are optional.

superposition: adding displacements or fields at the same place and time

interference: the pattern produced by overlapping waves

displacement: change from initial position to final position, with direction

01 Bring back an earlier idea

Across a stationary boundary, frequency remains 3 Hz while speed changes from 15 to 21 m/s.
Find both wavelengths.

Word help: boundary = the place where a medium or constraint changes

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $y_{resultant} = y_1 + y_2$.
Add signed displacements at the same place and time. Local cancellation does not destroy all wave energy.

Worked reasoning: At one point, two pulses have displacements +3 cm and -1 cm. Superposition gives +2 cm there. Add signed displacements, not their absolute values.

Your turn: Draw the two component displacements and the resultant at that point.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Represent and interpret

Two upward pulses overlap.
Sketch the combined displacement. Then sketch equal upward/downward pulses overlapping.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Sketch and label your response.

Marzano check - circle: ① ② ③

04 Solve and justify

Two coherent sinusoidal waves at a point have equal amplitude A.
Compare total amplitude for phase differences 0° and 180°.
Explain what must remain stable for persistent interference.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

Two rope pulses momentarily cancel. A student says the pulses are gone forever. What happens after they pass in an ideal linear medium?

Word help: ideal = using stated simplifying conditions

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

06 Choose a method

A process has ΔU=-24 J and W_{by}=32 J.
Find Q and decide whether heat enters or leaves.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

07 Exit check

Three instantaneous displacements are +5,-2,-6 mm. **Find** resultant and state whether this single value determines all wave energy.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Two waves at a point have equal amplitudes 4 cm. **Find** maximum constructive amplitude and complete destructive amplitude; explain the required relative phases.

Sentence starter: Under the assumption …, the evidence … supports … because … .

Marzano check - circle: 1 2 3